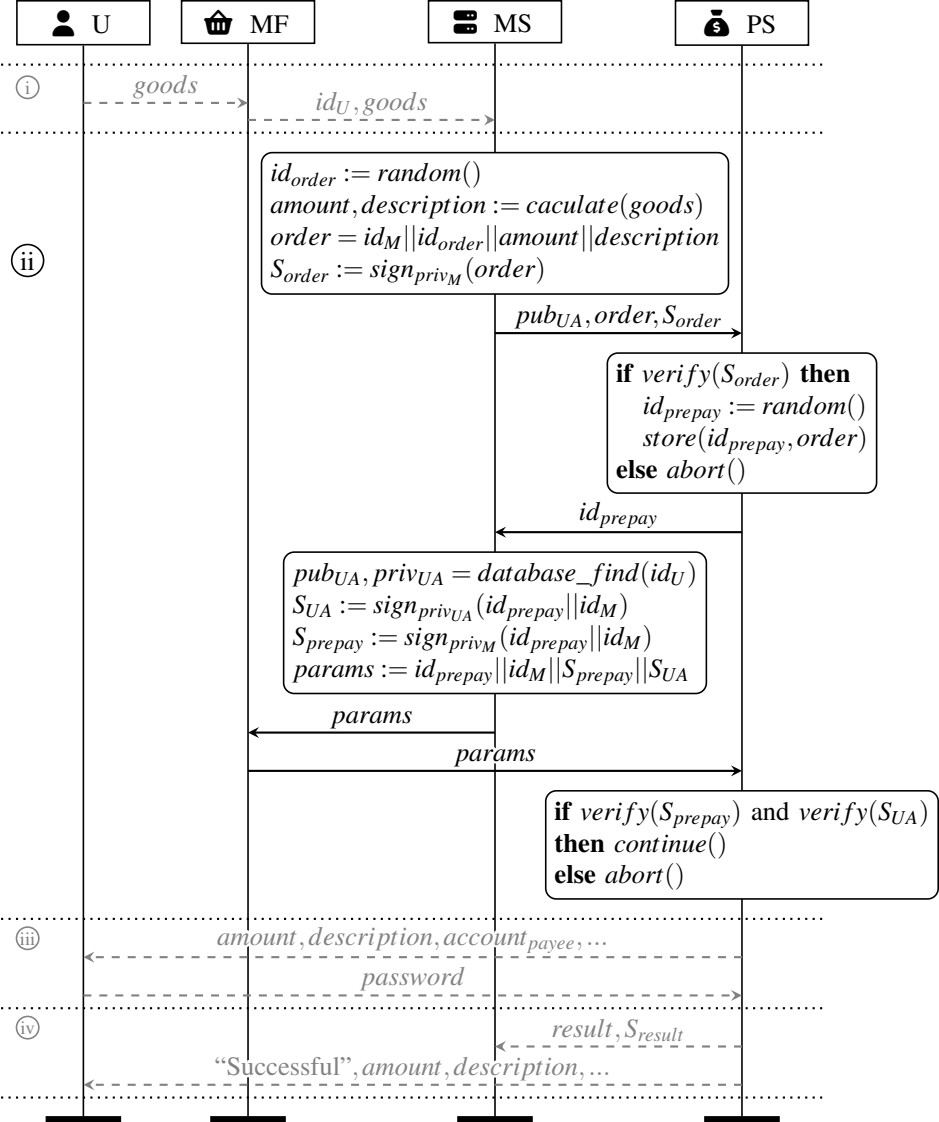

U
MF
MS
PS
i
goods
$id_U, goods$
ii
$id_{order} := random()$
$amount, description := caculate(goods)$
$order = id_M || id_{order} || amount || description$
$S_{order} := sign_{priv_M}(order)$
$pub_{UA}, order, S_{order}$
if $verify(S_{order})$ **then**
$id_{prepay} := random()$
$store(id_{prepay}, order)$
else $abort()$
id_{prepay}
$pub_{UA}, priv_{UA} = database_find(id_U)$
$S_{UA} := sign_{priv_{UA}}(id_{prepay} || id_M)$
$S_{prepay} := sign_{priv_M}(id_{prepay} || id_M)$
$params := id_{prepay} || id_M || S_{prepay} || S_{UA}$
$params$
$params$
if $verify(S_{prepay})$ and $verify(S_{UA})$
then $continue()$
else $abort()$
iii
$amount, description, account_{payee}, ...$
$password$
iv
$result, S_{result}$
"Successful", $amount, description, ...$